THE BOCKSTEIN BRIDGE

The Functor Connecting the Nontrivial Algebraic Bundle to Twistor Space

March 12, 2026

The Result

The nontrivial algebraic bundle $0 \to \mathbb{Z} \to \mathbb{Q} \to \mathbb{Q}/\mathbb{Z} \to 0$, proved to be non-splitting in "The Bundle Structure of the Integers," is IDENTICAL to the degree-2 Bockstein exact sequence of projective twistor space CP^3.

The functor is: apply the coefficient sequence $0 \to \mathbb{Z} \to \mathbb{Q} \to \mathbb{Q}/\mathbb{Z} \to 0$ to the singular cohomology of CP^3, and read the long exact sequence at degree 2.

This is not a construction. This is a computation. The bridge was already there.

The Computation

Step 1: Cohomology of CP^3 with integer coefficients

The cohomology ring of CP^3 is $H^*(CP^3; \mathbb{Z}) = \mathbb{Z}[y]/(y^4)$, where $y \in H^2(CP^3; \mathbb{Z})$ is the hyperplane class. Explicitly:

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H⁰(CP³; ℤ) = ℤ
H¹(CP³; ℤ) = 0
H²(CP³; ℤ) = ℤ    ← generated by y (the hyperplane class)
H³(CP³; ℤ) = 0
H⁴(CP³; ℤ) = ℤ    ← generated by y²
H⁵(CP³; ℤ) = 0
H⁶(CP³; ℤ) = ℤ    ← generated by y³ (the fundamental class)
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All groups are free abelian. No torsion.

Step 2: The first Chern class isomorphism

Since $H^1(CP^3; \mathbb{Z}) = 0$, the exponential sheaf sequence $0 \to \mathbb{Z} \to O \to O^* \to 0$ gives

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c₁: Pic(CP³) = H¹(CP³, O*) →≅ H²(CP³; ℤ) = ℤ
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This is an ISOMORPHISM. Every line bundle $O(n)$ on CP^3 is completely classified by its first Chern class $c_1(O(n)) = n \in \mathbb{Z}$.

The Penrose transform uses $H^1(PT, O(n))$ for various n. The integer n — the twist — lives in $H^2(CP^3; \mathbb{Z}) = \mathbb{Z}$ via the Chern class.

Step 3: Cohomology with $\mathbb{Q}$ and $\mathbb{Q}/\mathbb{Z}$ coefficients

Since all $H^k(CP^3; \mathbb{Z})$ are free:

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H^k(CP³; ℚ) = H^k(CP³; ℤ) ⊗ ℚ
H^k(CP³; ℚ/ℤ) = H^k(CP³; ℤ) ⊗ ℚ/ℤ
```

(No Tor terms because all integer cohomology groups are torsion-free.)

At degree 2:
$H^2(CP^3; \mathbb{Z}) = \mathbb{Z}$
$H^2(CP^3; \mathbb{Q}) = \mathbb{Q}$
$H^2(CP^3; \mathbb{Q}/\mathbb{Z}) = \mathbb{Q}/\mathbb{Z}$

Step 4: The Bockstein long exact sequence

The coefficient sequence $0 \to \mathbb{Z} \to \mathbb{Q} \to \mathbb{Q}/\mathbb{Z} \to 0$ induces:

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··· → H^n(CP³; ℤ) → H^n(CP³; ℚ) → H^n(CP³; ℚ/ℤ) →β H^{n+1}(CP³; ℤ) → ···
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At degree $1 \to 2$: The Bockstein $\beta: H^1(CP^3; \mathbb{Q}/\mathbb{Z}) \to H^2(CP^3; \mathbb{Z})$ is the zero map (since $H^1(CP^3; \mathbb{Q}/\mathbb{Z}) = 0$).

At degree $2 \to 3$: The Bockstein $\beta: H^2(CP^3; \mathbb{Q}/\mathbb{Z}) \to H^3(CP^3; \mathbb{Z})$ is the zero map (since $H^3(CP^3; \mathbb{Z}) = 0$).

Therefore the degree-2 portion of the long exact sequence is:

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0 → H²(CP³; ℤ) → H²(CP³; ℚ) → H²(CP³; ℚ/ℤ) → 0
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which is:

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║                                       ╕
│                                       │
│        0 → ℤ → ℚ → ℚ/ℤ → 0            │
│                                       │
│  THE SAME EXACT SEQUENCE.             │
│  NON-SPLITTING INHERITED.             │
│  THE BUNDLE OF THE INTEGERS           │
│  IS THE DEGREE-2 BOCKSTEIN            │
│  OF TWISTOR SPACE.                    │
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Step 5: Reading the three terms physically

FIBER: $H^2(CP^3; \mathbb{Z}) = \mathbb{Z} = \mathrm{Pic}(CP^3)$
This classifies line bundles O(n).
The integer n determines helicity via $h = -(n+2)/2$.
This is the "no return" part — the integers, torsion-free.

TOTAL: $H^2(CP^3; \mathbb{Q}) = \mathbb{Q}$
The rational extension. The continuum.
Contains the integers (line bundles) as a subgroup.
Cannot be retracted back to $\mathbb{Z}$.

BASE: $H^2(CP^3; \mathbb{Q}/\mathbb{Z}) = \mathbb{Q}/\mathbb{Z} = \mathrm{SBr}(CP^3)$
The special Brauer group. Classifies gerbes (higher bundles).
Contains all finite cyclic groups — all torsion.
Every element has finite order — guaranteed return.

The non-splitting says: you cannot separate the line bundle classification (helicities) from the Brauer group (gerbes/torsion) by any cohomological operation. They are bound in the rational cohomology, which cannot be split back.

Step 6: Universality

This works for ANY CP^n, not just CP^3:

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0 → H²(CP^n; ℤ) → H²(CP^n; ℚ) → H²(CP^n; ℚ/ℤ) → 0
=   0 → ℤ → ℚ → ℚ/ℤ → 0
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for all $n \geq 1$. The bundle structure is universal across all projective spaces.

It also works for any space X with $H^1(X; \mathbb{Z}) = 0$ and $H^2(X; \mathbb{Z}) = \mathbb{Z}$ and $H^3(X; \mathbb{Z}) = 0$. The Bockstein sequence at degree 2 always produces the same non-splitting exact sequence.

Step 7: The prime-by-prime structure on twistor space

Since $H^2(CP^3; \mathbb{Q}/\mathbb{Z}) = \mathbb{Q}/\mathbb{Z} = \bigoplus_p \mathbb{Z}(p\infty)$, the Prüfer decomposition applies directly to the Brauer-like invariants of twistor space:

$$H^2(CP^3; \mathbb{Q}/\mathbb{Z}) = \mathbb{Z}(2\infty) \oplus \mathbb{Z}(3\infty) \oplus \mathbb{Z}(5\infty) \oplus \mathbb{Z}(7\infty) \oplus \cdots$$

One tower per prime. On twistor space. The same towers we studied in "The Bundle Structure of the Integers." They live here too.

And the distortion — measured by $\mathrm{Ext}^1(\mathbb{Q}/\mathbb{Z}, \mathbb{Z}) \cong \hat{\mathbb{Z}} = \prod_p \mathbb{Z}_p$ — is the same profinite completion. One twist per prime. The continuum lives in the binding.

Summary

THE FUNCTOR: The Bockstein homomorphism associated to $0 \to \mathbb{Z} \to \mathbb{Q} \to \mathbb{Q}/\mathbb{Z} \to 0$

THE INPUT: Twistor space CP^3 (or any projective space CP^n)

THE OUTPUT: At degree 2, the exact sequence $0 \to \mathbb{Z} \to \mathbb{Q} \to \mathbb{Q}/\mathbb{Z} \to 0$

THE READING:
$\mathbb{Z}$ = $\mathrm{Pic}(CP^3)$ = line bundles = helicities
$\mathbb{Q}$ = rational cohomology = the continuum binding
$\mathbb{Q}/\mathbb{Z}$ = $\mathrm{SBr}(CP^3)$ = special Brauer group = torsion/gerbes = the discrete

THE BRIDGE: The algebraic bundle of the integers and the twistor space line bundle classification are the same mathematical object, connected by the Bockstein functor at degree 2.

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